

Properties of Natural and Made Materials

Name: _____

Date: _____

Year 4 Science — Chemical Sciences

ACARA v9.0: AC9S4U04

Learning Intention

I am learning to examine the properties of natural and made materials and consider how these properties influence their use.

Success Criteria

- I can identify whether a material is natural or made (processed).
- I can describe properties of common materials.
- I can explain why a material's properties make it suitable for a particular use.

Natural or Made? Think About Origin

Natural materials come directly from nature (wood, cotton, wool). Made materials are processed or manufactured from natural materials (plastic, glass, some metals like steel).

Worked Example

Why are raincoats made from plastic, not wool?

Plastic is waterproof and flexible.

Wool absorbs water and would get heavy and wet.

Part A — Warm Up

Circle the natural material: **plastic** **cotton** **glass**

True or False: All made materials come from natural materials originally. _____

Name one property of glass (e.g. see-through, hard). _____

Part B — Core Skills

Sort into Natural or Made:

wool, plastic, wood, steel, cotton, glass

Why is metal used for cooking pots instead of wood?

List 2 properties that make rubber good for shoe soles.

1. _____ 2. _____

Why might a company choose plastic instead of glass for a kids' water bottle?

Word Bank

natural • made • property • fibre • metal • flexible • waterproof • durable • transparent

Working Space — Draw 3 Natural Materials and 3 Made Materials

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Part C — Challenge: The School Uniform Investigation

A Geelong primary school is deciding on new winter jumpers. One option is 100% wool, and another is a made (synthetic) polyester fleece material.

Q8. Name one property of wool that would make it a good choice for a winter jumper.

Q9. Name one property of polyester fleece that might make it a good choice instead.

Q10. The school decides to choose the material that dries fastest after getting wet in the rain, since students walk to school. Which material would you recommend, and explain your reasoning using material properties.

Answer Key — Part A

A1: cotton

A2: True — even made materials like plastic are originally processed from natural resources such as oil.

A3: Transparent/see-through, hard, smooth, or breakable (accept any reasonable property).

Answer Key — Part B

B1: Natural: wool, wood, cotton. Made: plastic, steel, glass.

B2: Metal can withstand high heat without melting or burning, unlike wood, making it safe for cooking.

B3: Grippy/good friction on surfaces and flexible/bends easily (accept similar reasonable properties).

B4: Plastic is lighter, does not shatter/break as easily as glass, making it safer for children to use.

Answer Key — Part C

C8: Wool is warm and naturally insulating, helping keep students warm in winter.

C9: Polyester fleece is often lighter, quick-drying, and less likely to shrink or hold moisture compared to wool (accept similar reasonable properties).

C10: Polyester fleece — synthetic fleece generally dries faster than wool because wool fibres absorb and hold onto water, while polyester fibres absorb very little water and let it evaporate more quickly.

I Can Checklist

☐ I can identify natural vs made materials.

☐ I can explain why properties suit a material's use.

☐ I can describe material properties.